

الجمهورية الجزائرية الديمقراطية الشعبية

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وزارة التعليم العالي والبحث العلمي

ⵎⵓⵏⵉⵙⵜ ⵉⵏⵉⵎⵙⵉⵎ ⵉⵏⵉⵎⵙⵉⵎ ⵉⵏⵉⵎⵙⵉⵎ

Ministère de l'Enseignement Supérieur et de la Recherche Scientifique



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المدرسة الوطنية العليا للإعلام الآلي
ⵎⵓⵏⵉⵙⵜ ⵉⵏⵉⵎⵙⵉⵎ ⵉⵏⵉⵎⵙⵉⵎ ⵉⵏⵉⵎⵙⵉⵎ
École nationale Supérieure d'Informatique

SSR, SIQ2

LAB7C: SH DETECTION

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1. Session Hijacking Overview:

Session hijacking attacks target active TCP/IP communication sessions. The attacker intercepts traffic between two systems and may steal session information or monitor user activity.

2. Session Hijacking Launching:

First we launch Wireshark in the victim machine (Windows 10), next, we start Bettercap in the attacker machine (Parrot7):

```
bettercap -iface ens33
```

3. Network Reconnaissance and Probing:

Inside Bettercap, several modules were activated to discover hosts and monitor network activity. First, we enabled probing with:

```
net.probe on
```

This module sends probe packets to hosts in the subnet to identify active systems.

Then we enabled network reconnaissance using:

```
net.recon on
```

This module reads the ARP table and continuously detects active IP addresses on the network. Bettercap started listing hosts detected in the subnet in real time.

```
[root@parrot]~# #bettercap -iface ens33
bettercap v2.33.0 (built for linux amd64 with go1.24.4) [type 'help' for a list of commands]

10.10.0.0/24 > 10.10.0.130 » [22:28:13] [sys.log] [war] Could not find mac for
10.10.0.0/24 > 10.10.0.130 » net.probe on
10.10.0.0/24 > 10.10.0.130 » [22:30:23] [sys.log] [inf] net.probe probing 256 addresses on 10.10.0.0/24
10.10.0.0/24 > 10.10.0.130 » [22:30:23] [sys.log] [inf] net.probe starting net.recon as a requirement for net.probe
10.10.0.0/24 > 10.10.0.130 » [22:30:23] [endpoint.new] endpoint 10.10.0.129 detected as 00:0c:29:89:26:c5 (VMware, Inc.).
10.10.0.0/24 > 10.10.0.130 » [22:30:23] [endpoint.new] endpoint 10.10.0.2 detected as 00:50:56:f6:a0:c9 (VMware, Inc.).
10.10.0.0/24 > 10.10.0.130 » [22:30:23] [endpoint.new] endpoint 10.10.0.1 detected as 00:50:56:c0:00:02 (VMware, Inc.).
10.10.0.0/24 > 10.10.0.130 » [22:30:26] [endpoint.new] endpoint 10.10.0.254 detected as 00:50:56:f6:85:ad (VMware, Inc.).
10.10.0.0/24 > 10.10.0.130 » net.recon on
10.10.0.0/24 > 10.10.0.130 » [22:30:45] [sys.log] [err] module net.recon is already running
10.10.0.0/24 > 10.10.0.130 » net.sniff on
10.10.0.0/24 > 10.10.0.130 » [22:31:05] [net.sniff.mdns] mdns CLIENT1 : Unknown query for CLIENT1.local
10.10.0.0/24 > 10.10.0.130 » [22:31:05] [net.sniff.mdns] mdns CLIENT1 : CLIENT1.local is 10.10.0.129
10.10.0.0/24 > 10.10.0.130 »
```

4. Sniffing Network Traffic:

After reconnaissance, packet sniffing was enabled:

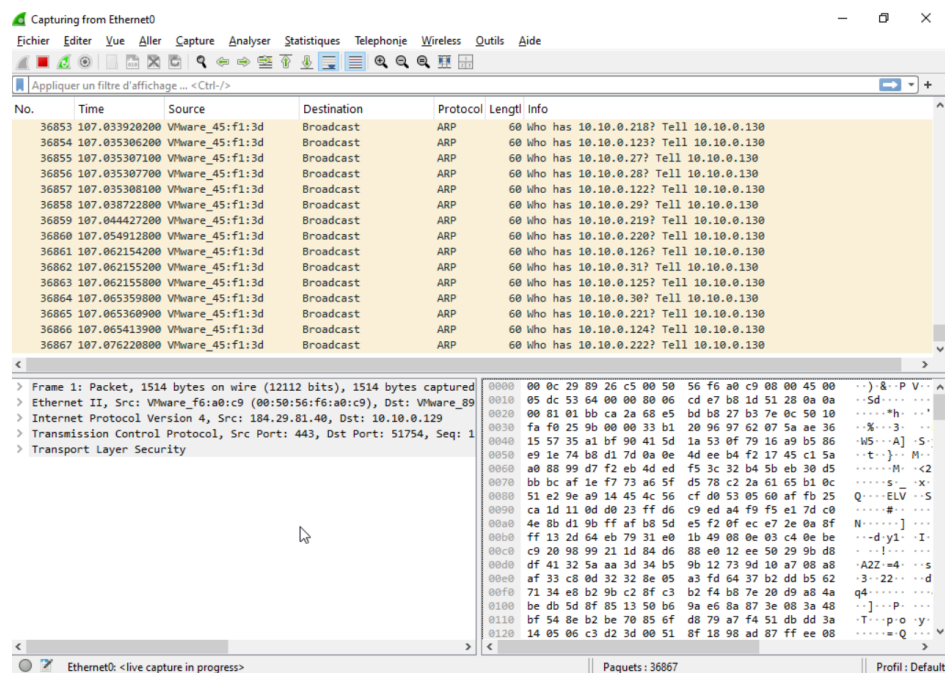
net.sniffon

This module allowed Bettercap to monitor and sniff network traffic between systems in the network. Once enabled, Bettercap started intercepting packets from multiple machines connected to the subnet.

5. Detecting the Attack with Wireshark:

We then switched back to the victim machine and observed the traffic captured by Wireshark.

Wireshark showed a very large number of ARP packets being generated from the attacker machine (10.10.0.130). These ARP broadcast requests indicated suspicious behavior because the attacker machine was communicating with many IP addresses in the subnet.



The captured traffic showed that packets from the victim machine were first redirected to the attacker machine before reaching the gateway. Similarly, incoming packets destined for the victim were also forwarded through the attacker system.

This behavior confirmed that the attacker machine was acting as an intermediary between the victim and the network, which is a typical characteristic of session hijacking and Man-in-the-Middle attacks.